AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_E2F4C22098
Taxa	Brachiosaurus
Clade	Sauropod
PBDB Records	20
Mass (regression)	48781.1 kg (Campione & Evans 2012)
Mass (volumetric)	29029.0 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	34.558 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6188 (partial)
Axiom I Entropy	0.9471
Axiom II λ■	0.1805 (symmetry 100%)
Axiom III Uniformity	0.3816
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	66.7% Measured / 33.3% Inferred
Elements	27
CoG (m)	[2.684, 4.324, 0.0]
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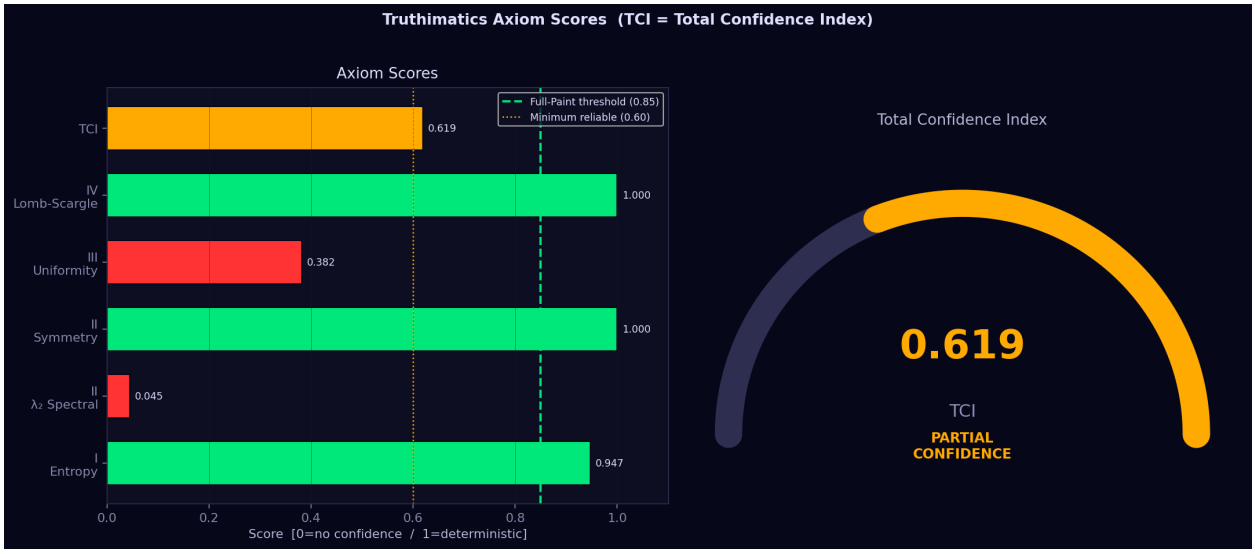
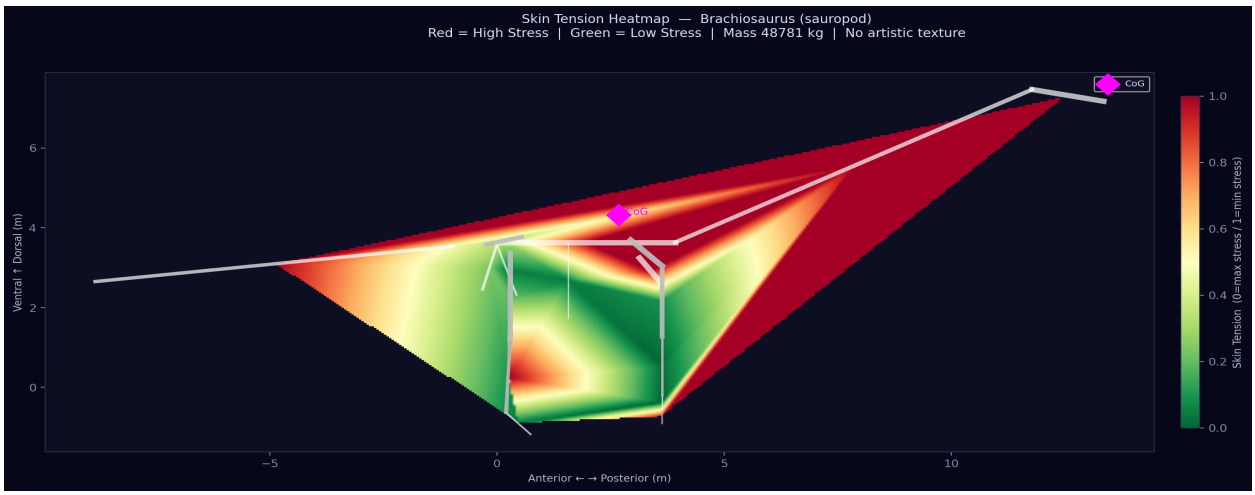
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

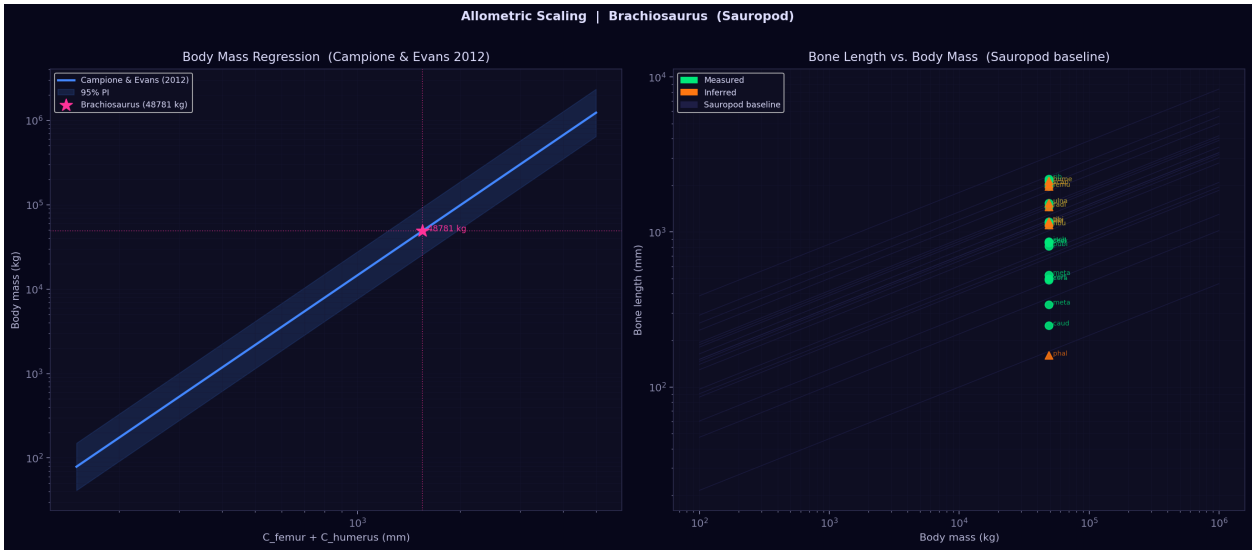
Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
caudal_vertebra	2.608	4.573	7.181	15.000	OK
cervical_vertebra	1.875	9.475	11.350	14.978	OK
coracoid	1.032	2.231	3.263	15.000	OK
femur	4.035	40.896	44.931	3.784	WARNING
fibula	0.000	0.000	0.000	15.000	OK
humerus	5.451	69.802	75.253	2.259	WARNING
ilium	4.919	25.789	30.708	5.536	OK
ischium	5.060	34.401	39.460	4.308	OK
metacarpal	51.241	260.148	311.389	0.546	CRITICAL
metatarsal	33.476	84.399	117.875	1.442	CRITICAL
phalanx	0.000	0.000	0.000	15.000	OK
pubis	9.917	87.886	97.803	1.738	CRITICAL
radius	8.196	148.771	156.967	1.083	CRITICAL
rib	1.561	102.480	104.041	1.634	CRITICAL
scapula	0.652	4.664	5.316	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	11.794	58.290	70.083	2.426	WARNING
ulna	6.887	120.870	127.757	1.331	CRITICAL
vertebra	2.767	5.913	8.680	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	3.661 m	Femur + tibia + metatarsal
Estimated Speed	15.5 km/h (4.31 m/s)	Gait: Walk | Fr=0.516
Stride Length	6.888 m	Alexander (1976) / Thulborn (1990)
Bite Force	4521.8 N (460.9 kgf)	Sauropod regression (Whitlock 2011)
Skull Length	860.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4239	Circumference / length (dimensionless)
Limb Strength Index	2.7e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	27.13 m	Allometric from femur length
Est. Neck Length	11.39 m	Allometric from femur length
Est. Tail Length	11.39 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
caudal_vertebra	250.0	195.0	3807.76	4753.2	Measured	0.88	15.00
cervical_vertebra	850.0	230.0	18010.87	22483.0	Measured	0.88	14.98
coracoid	490.0	310.0	18861.64	23545.0	Measured	0.88	15.00
femur	1964.0	265.0	55245.04	68962.4	Measured	0.88	3.78
femur_right	1964.0	265.0	55245.04	68962.4	Symmetry	0.84	3.78
fibula	1110.0	65.0	1326.00	1390.7	Measured	0.88	15.00
fibula_right	1110.0	65.0	1326.00	1390.7	Symmetry	0.84	15.00
humerus	2135.0	228.0	44455.72	55494.1	Measured	0.88	2.26
humerus_right	2135.0	228.0	44455.72	55494.1	Symmetry	0.84	2.26
ilium	1150.0	240.0	26532.63	33120.7	Measured	0.88	5.54
ilium_right	1150.0	240.0	26532.63	33120.7	Symmetry	0.84	5.54
ischium	870.0	140.0	6830.23	8526.2	Measured	0.88	4.31
metacarpal	340.0	74.0	277.83	228.6	Measured	0.88	0.55
metatarsal	530.0	92.0	1796.85	2243.0	Measured	0.88	1.44
phalanx	160.0	99.9	192.80	149.3	Constraint	0.40	15.00
pubis	810.0	100.0	3244.48	4050.1	Measured	0.88	1.74
radius	1460.0	110.0	7076.17	8833.2	Measured	0.88	1.08
radius_right	1460.0	110.0	7076.17	8833.2	Symmetry	0.84	1.08
rib	2200.0	55.0	2665.68	3327.6	Measured	0.88	1.63
scapula	2040.0	390.0	124285.21	155145.2	Measured	0.88	15.00
scapula_right	2040.0	390.0	124285.21	155145.2	Symmetry	0.84	15.00
skull	860.0	480.0	79367.19	99074.1	Measured	0.88	15.00
tibia	1167.0	155.0	11230.38	14018.9	Measured	0.88	2.43
tibia_right	1167.0	155.0	11230.38	14018.9	Symmetry	0.84	2.43
ulna	1540.0	120.0	8882.66	11088.2	Measured	0.88	1.33
ulna_right	1540.0	120.0	8882.66	11088.2	Symmetry	0.84	1.33
vertebra	500.0	320.0	20508.32	25600.5	Measured	0.88	15.00